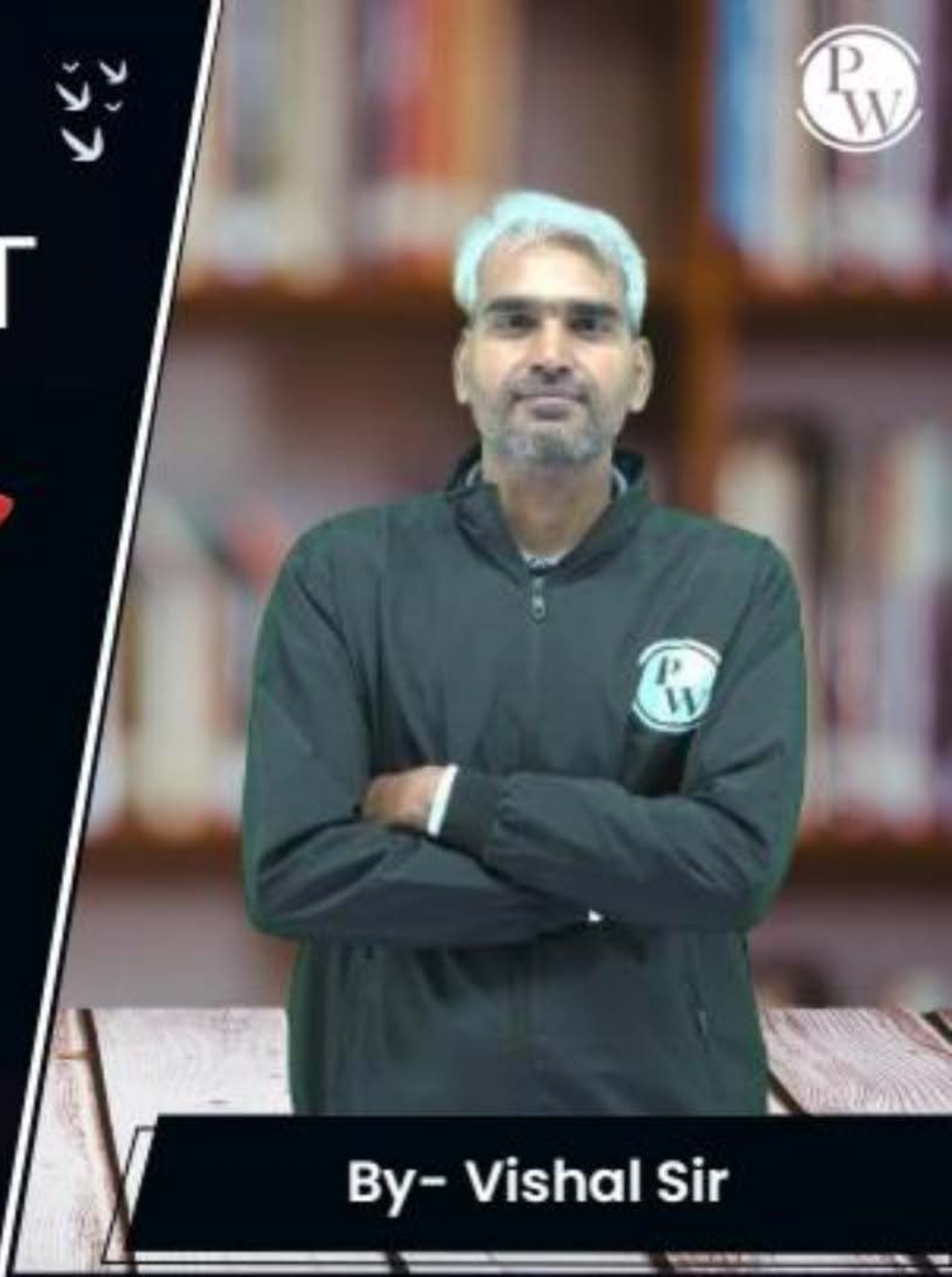
Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



Topic Grammar and Types of grammar

Topic Types of Derivation Tree

Slide

Topics to be Covered



Topic Types of Parser

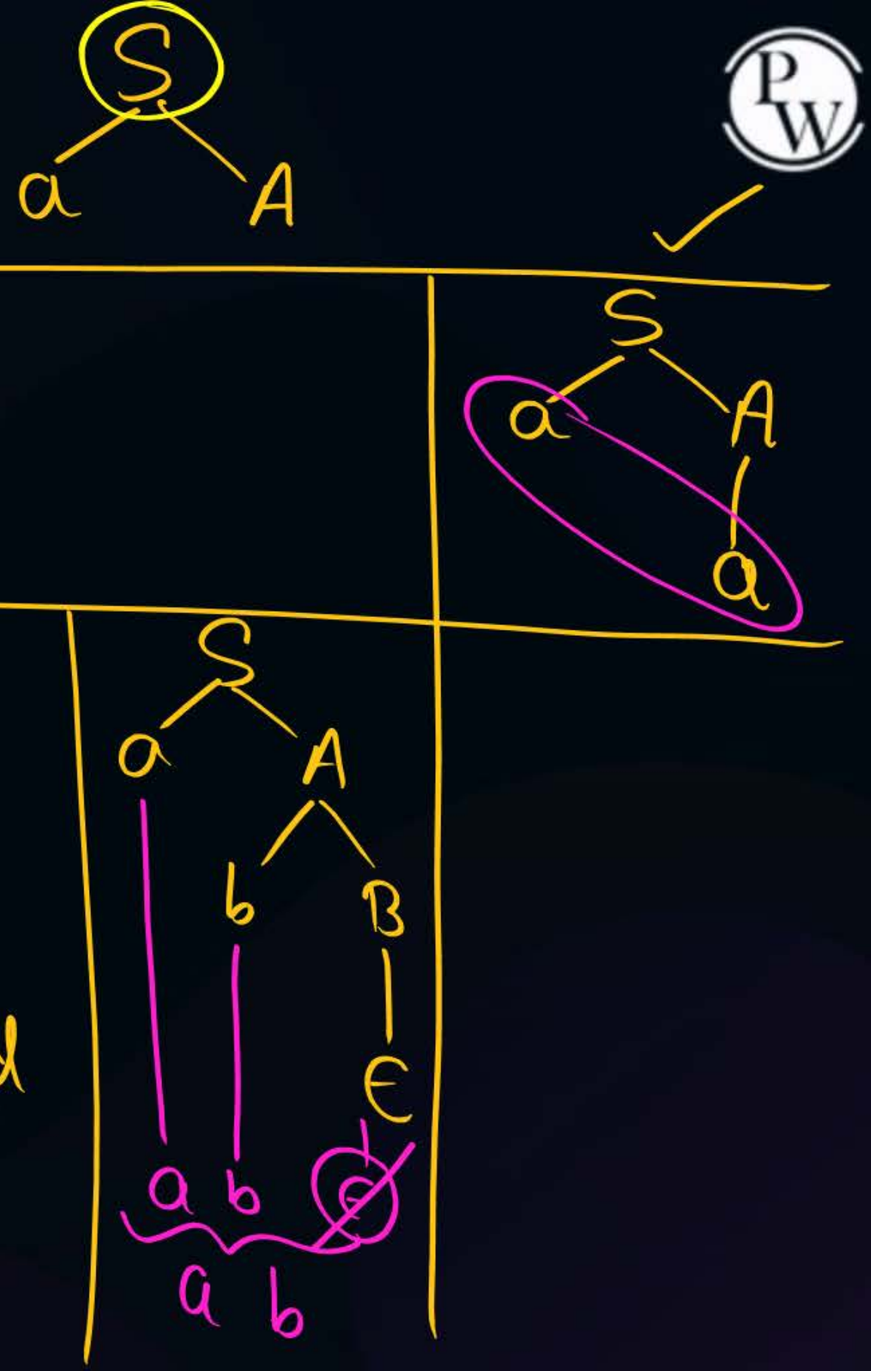
Topic Elimination of Left Recursion

Topic Left Factoring

H.W.

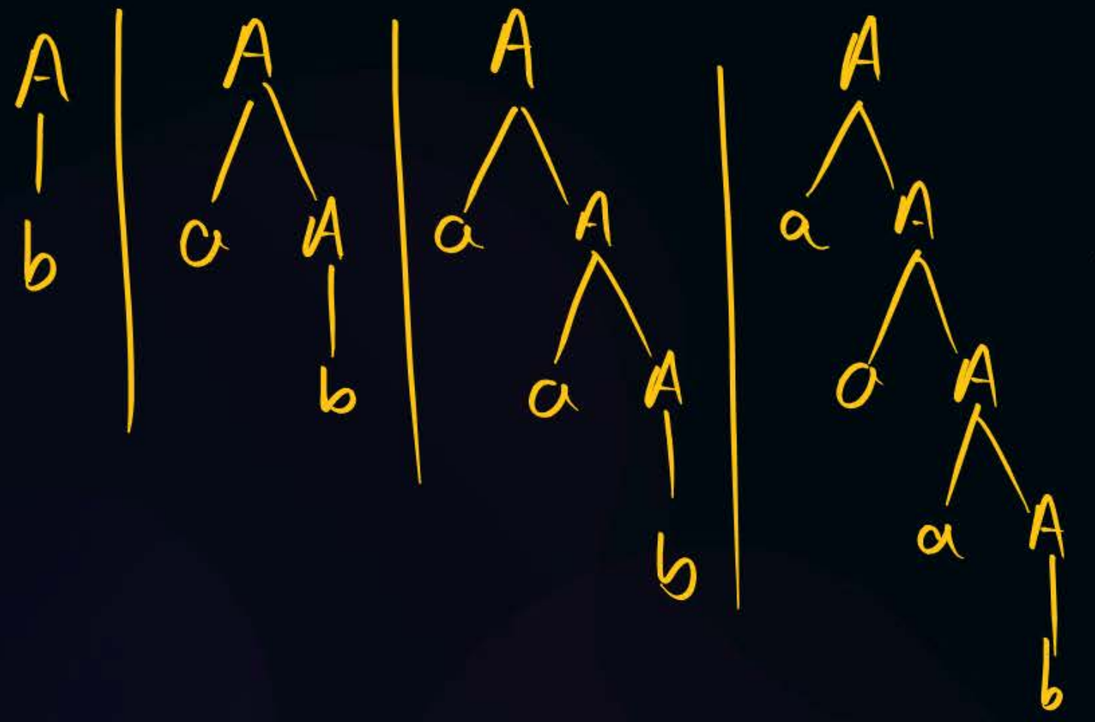
$$G: \begin{cases} S \to aA & \checkmark \\ A \to bB/a \\ B \to Cd/\epsilon \\ C \to xxd/ddx \end{cases}$$

Language = {aa, ab, abccdd, abddxd}



Recursive grammar :-

eg: $A \rightarrow aA/b \Rightarrow L = \{b, ab, aab, aaab, aaaab, \ldots\}$



a^* ← ϵ included

$a^* \equiv$ any number of a including '0' {i.e. ϵ}

$a^* = \epsilon, a, aa, aaa, \ldots$ (R.E.)

ϵ not allowed.

$a^+ \equiv$ At least one 'a'

$a^+ = \{a, aa, aaa, \ldots\}$ (R.E.)

- **Regular Expression :-**

$(a.b)^* = \{\epsilon, ab, abab, ababab, \ldots\}$

$(a+b) = \{a, b\}$

$(a+b)^* = \{\epsilon, a, b, ab, ba, aaa, bbb, \ldots\}$

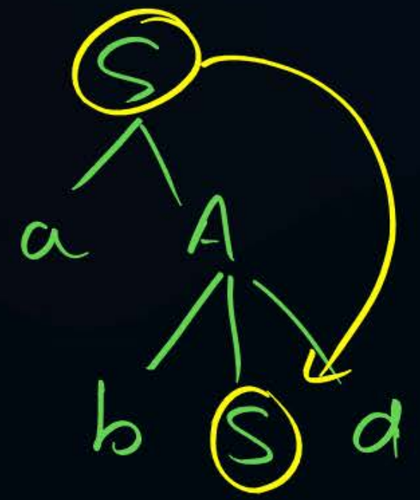
Any combination of strings of a & b including ϵ

$ab\,c^*$ = {ab, abc, abcc, abccc, ...}

✡ Indirect Recursion

$S \rightarrow aA \mid b$

$A \rightarrow bSd \mid c$



S derived from S, but indirectly

∴ Indirect Recursion



Topic : Types of Parser

These are two types of Parser

- Top-down Parser
  - further classification will be there in upcoming Sessions
- Bottom-Up parser
  - Same

Slide



Topic : Types of Parser

Samajh aaya

Yes

No

Top-down Parser :- If any parser uses top-down approach for parsing, then it uses leftmost derivation in the construction of parse tree.

→ Top-down parser will start the construction of Parse tree starting from the Start symbol.

→ Top-down parser can see only one symbol of input string at a time from left to right.

Slide



Topic : Types of Parser

Bottom-up Parser : → If any parser uses bottom up approach for the purpose of parsing, then it will use reverse of rightmost derivation.

→ Bottom up parser will start the construction of parse tree from given input string, and it will try to reduce it into Start symbol.

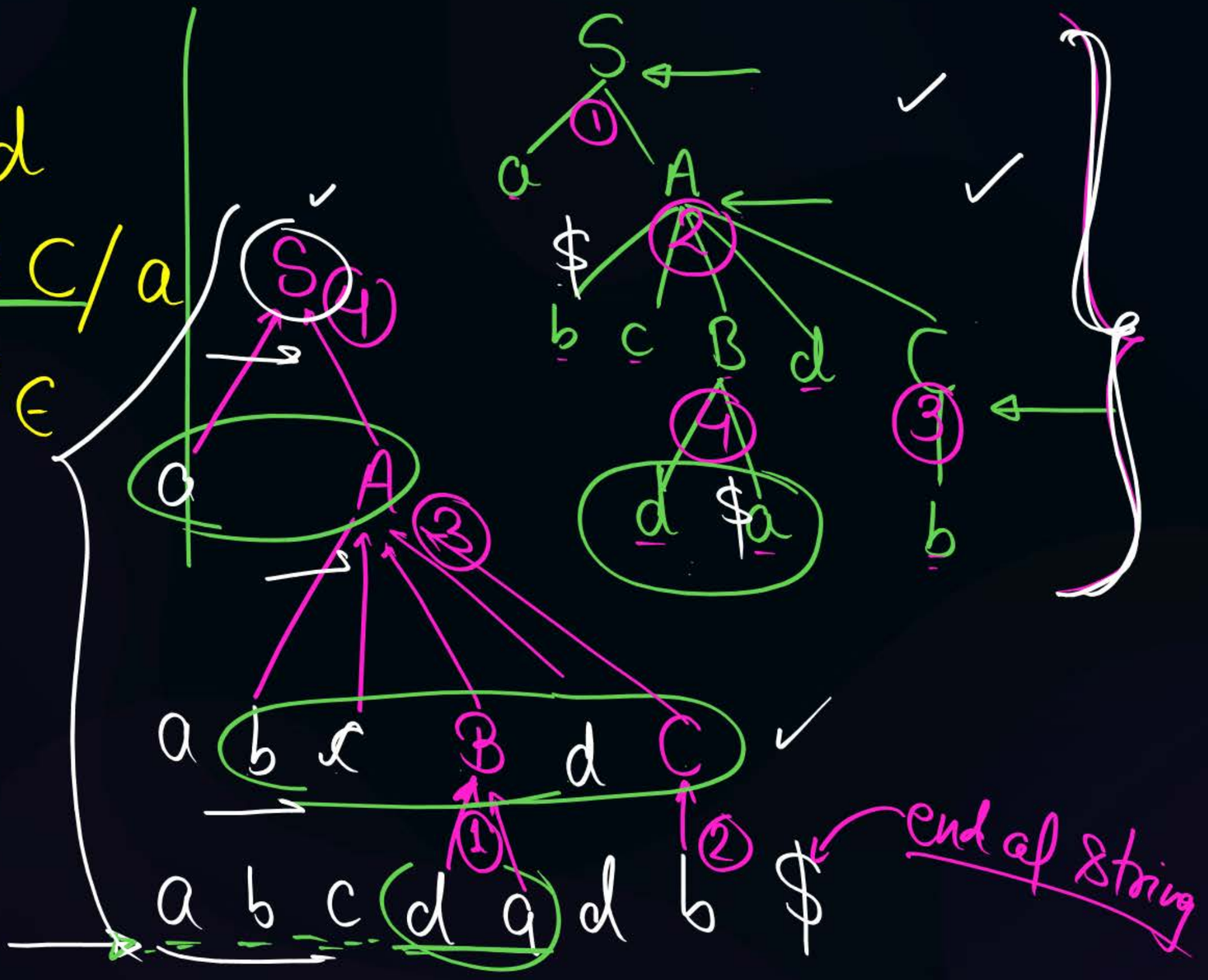
Slide

$S \rightarrow aA / d$

$A \rightarrow bcBdC / a$

$B \rightarrow da / \epsilon$

$C \rightarrow b$





Topic : Recursive Contex Free Grammar(CFG)

L.R & R.R

Yes

No

Context free grammar Can be

Non Recursive

(or)

Recursive

$S \rightarrow aA/b$ { No Recursion }

eg: $S \rightarrow aSbs/bsaS/\epsilon$

eg: $A \rightarrow A\alpha/\beta$ { Left Recursive }

$A \rightarrow \alpha A/\beta$ { Right Recursive }

$A \rightarrow \alpha A\beta/\gamma$ { Middle Recursion }

Slide

Topic : Problem with Left Recursive CFG

* Top-down parse will get stuck into an infinite loop w.r.t. Left Recursive grammar

eg grammar = $\overbrace{A \rightarrow A\alpha / \alpha}^{\text{Left Recursive}}$

$L = \{\alpha, \alpha\alpha, \ldots\} = \{\alpha^{+}\}$

let input string = $\alpha^3 = \alpha\alpha\alpha$



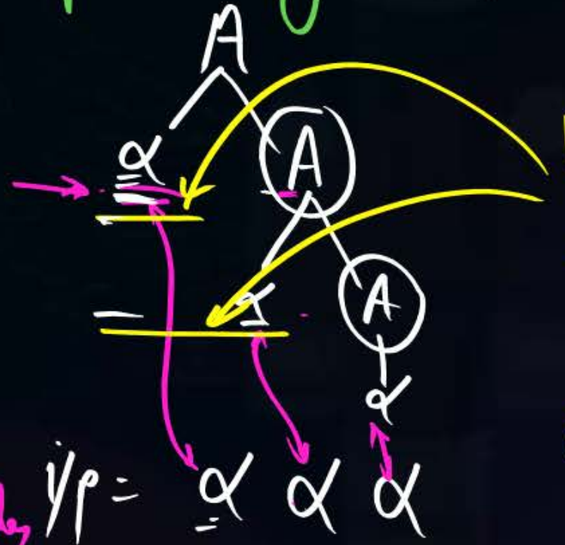
i/p = ααα

As soon as first non-terminal is encountered from L.H.S it will try to expand it will never know how many α it has already produced

grammar = $\overbrace{A \rightarrow \alpha A / \alpha}^{\text{Right Recursive}}$

language = $\{\alpha, \alpha\alpha, \alpha\alpha\alpha, \ldots\} = \{\alpha^{+}\}$

let input string = $\alpha^3 = \alpha\alpha\alpha$



i/p = ααα

While scanning from left to Right we know that how many α we have already produced.

Slide

Topic : Elimination of Left Recursion

$A \rightarrow A\alpha / \beta$ — Left Recursion

$\equiv$ Equivalent grammar

$L = \{\beta, \beta\alpha, \beta\alpha\alpha, \beta\alpha\alpha\alpha, \ldots\}$

$L = \{\beta.(\epsilon + \alpha + \alpha\alpha + \alpha\alpha\alpha + \ldots)\}$

$L = \beta.\alpha^*$

Yes / No

Language $\Rightarrow L = \beta\alpha^*$

try to write Right recursive language for this gramman

let α^* is generated Right recursively using A'

$L = \beta\alpha^*$

Let α^* generated using A'

$\therefore$ grammar G' =

$A \rightarrow \beta A'$

$A' \rightarrow \alpha A' / \epsilon$

{ i.e. Productions of A' will be $A' \rightarrow \alpha A' / \epsilon$ }

Right Recursion

Slide

α^*

$A \rightarrow A\alpha / \epsilon$ $\Rightarrow$ $L = ?$ α^*

$A' \rightarrow \alpha A' / \epsilon$ $\Rightarrow$ $L = ?$ α^*

$\equiv$

$A \rightarrow \alpha A / \epsilon$



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU